

T3 BSM UV and matched EFT contributions

EFT order: 5; loop order: 1. The EFT expressions show $\Delta\mathcal{L}_{\text{EFT}} = \mathcal{L}_{\text{matched}}^{\text{SM+BSM}} - \mathcal{L}_{\text{matched}}^{\text{SM}}$.

$$d_S = 2, \quad Y_S = \frac{1}{2}, \quad d_F = 1, \quad Y_F = 0$$

$$\text{BSM UV Lagrangian } \eta_{a_2} \bar{\epsilon}_{a_1, a_2} \overline{y_{N, i_1, r_1}} N_{r_1}^T C P_L \ell_{a_1, i_1}$$

$$+ \epsilon_{a_2, a_1} \eta_{a_1}^\dagger y_{N, i_1, r_1} \overline{\ell_{a_2, i_1}} P_R N_{r_1}$$

$$+ \frac{1}{2} \overline{\lambda_5} H_{a_1} H_{a_2} \eta_{a_1}^\dagger \eta_{a_2}^\dagger$$

$$+ \frac{1}{2} i N_{r_1}^T C \gamma^{\mu_1} D_{\mu_1} (N_{r_1})$$

$$- \frac{1}{2} M_N N_{r_1}^T C N_{r_1}$$

$$+ \frac{1}{2} \lambda_5 \eta_{a_1} \eta_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger$$

$$+ \lambda_3 H_{a_1} \eta_{a_2} H_{a_1}^\dagger \eta_{a_2}^\dagger$$

$$+ \lambda_4 \eta_{a_1} H_{a_2} H_{a_1}^\dagger \eta_{a_2}^\dagger$$

$$+ \frac{1}{2} \lambda_S \eta_{a_1} \eta_{a_2} \eta_{a_1}^\dagger \eta_{a_2}^\dagger$$

$$+ D_{\mu_1} (\eta_{a_1}) D_{\mu_1} (\eta_{a_1})^\dagger$$

$$+ M_\eta^2 (-\eta_{a_1}) \eta_{a_1}^\dagger$$

$$\Delta \mathcal{L}_{\text{EFT}} - \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_N^2} \right) \overline{y_{N, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{N, i_2, r_1} Y_{e, i_3, i_1} M_N^4}{4 (M_\eta^2 - M_N^2)^2}$$

$$- \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_N^2} \right) \overline{y_{N, i_1, r_1}} \overline{Y_{e, i_3, i_2}} \overline{e_{i_2}} P_L \ell_{a_1, i_1} y_{N, i_3, r_1} H_{a_1}^\dagger M_N^4}{4 (M_\eta^2 - M_N^2)^2}$$

$$- \frac{\hbar \overline{y_{N, i_1, r_1}} \overline{y_{N, i_2, r_1}} \overline{\lambda_5} \ell_{a_3, i_2}^T C P_L \ell_{a_1, i_1} H_{a_2} H_{a_4} \bar{\epsilon}_{a_1, a_2} \bar{\epsilon}_{a_3, a_4} M_N^3}{2 (M_\eta^2 - M_N^2)^2}$$

$$- \frac{\hbar \log \left(\frac{M_\eta^2}{M_N^2} \right) \overline{y_{N, i_1, r_1}} \overline{y_{N, i_2, r_1}} \overline{\lambda_5} \ell_{a_3, i_2}^T C P_L \ell_{a_1, i_1} H_{a_2} H_{a_4} \bar{\epsilon}_{a_1, a_2} \bar{\epsilon}_{a_3, a_4} M_N^3}{2 (M_\eta^2 - M_N^2)^2}$$

$$- \frac{\hbar \overline{\ell_{a_3, i_1}} C P_R \overline{\ell_{a_4, i_2}}^T y_{N, i_1, r_1} y_{N, i_2, r_1} \epsilon_{a_3, a_1} \epsilon_{a_4, a_2} \lambda_5 H_{a_1}^\dagger H_{a_2}^\dagger M_N^3}{2 (M_\eta^2 - M_N^2)^2}$$

$$- \frac{\hbar \log \left(\frac{M_\eta^2}{M_N^2} \right) \overline{\ell_{a_3, i_1}} C P_R \overline{\ell_{a_4, i_2}}^T y_{N, i_1, r_1} y_{N, i_2, r_1} \epsilon_{a_3, a_1} \epsilon_{a_4, a_2} \lambda_5 H_{a_1}^\dagger H_{a_2}^\dagger M_N^3}{2 (M_\eta^2 - M_N^2)^2}$$

$$- \frac{\hbar \overline{y_{N, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{N, i_2, r_1} Y_{e, i_3, i_1} M_N^2}{2 (M_\eta^2 - M_N^2)}$$

$$- \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_N^2} \right) \overline{y_{N, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} y_{N, i_2, r_1} Y_{e, i_3, i_1} M_N^2}{2 (M_\eta^2 - M_N^2)}$$

$$+ \frac{\hbar \overline{y_{N, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} M_\eta^2 y_{N, i_2, r_1} Y_{e, i_3, i_1} M_N^2}{4 (M_\eta^2 - M_N^2)^2}$$

$$+ \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_N^2} \right) \overline{y_{N, i_3, r_1}} \overline{\ell_{a_1, i_2}} P_R e_{i_1} H_{a_1} M_\eta^2 y_{N, i_2, r_1} Y_{e, i_3, i_1} M_N^2}{2 (M_\eta^2 - M_N^2)^2}$$

$$\begin{aligned}
& - \frac{\hbar \overline{y_{N,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{N,i_3,r_1} H_{a_1}^\dagger M_N^2}{2 (M_\eta^2 - M_N^2)} \\
& - \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_N^2} \right) \overline{y_{N,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{N,i_3,r_1} H_{a_1}^\dagger M_N^2}{2 (M_\eta^2 - M_N^2)} \\
& + \frac{\hbar \overline{y_{N,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_\eta^2 y_{N,i_3,r_1} H_{a_1}^\dagger M_N^2}{4 (M_\eta^2 - M_N^2)^2} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_N^2} \right) \overline{y_{N,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_\eta^2 y_{N,i_3,r_1} H_{a_1}^\dagger M_N^2}{2 (M_\eta^2 - M_N^2)^2} \\
& + \frac{\hbar \overline{y_{N,i_1,r_1}} \overline{y_{N,i_2,r_1}} \overline{\lambda_5 \ell_{a_3,i_2}}^T C P_L \ell_{a_1,i_1} H_{a_2} H_{a_4} M_\eta^2 \bar{\epsilon}_{a_1,a_2} \bar{\epsilon}_{a_3,a_4} M_N}{2 (M_\eta^2 - M_N^2)^2} \\
& + \frac{\hbar \overline{\ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_\eta^2 y_{N,i_1,r_1} y_{N,i_2,r_1} \epsilon_{a_3,a_1} \epsilon_{a_4,a_2} \lambda_5 H_{a_1}^\dagger H_{a_2}^\dagger M_N}{2 (M_\eta^2 - M_N^2)^2} \\
& - \frac{\hbar B_{\mu_1 \mu_2}^2}{24 \epsilon} \\
& - \frac{1}{24} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_\eta^2} \right) B_{\mu_1 \mu_2}^2 \\
& - \frac{\hbar \left(W_{\mu_1 \mu_2}^{A_1} \right)^2}{24 \epsilon} \\
& - \frac{1}{24} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_\eta^2} \right) \left(W_{\mu_1 \mu_2}^{A_1} \right)^2 \\
& + \frac{\hbar \overline{y_{N,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{N,i_2,r_1} Y_{e,i_3,i_1}}{4 \epsilon} \\
& - \frac{1}{8} \hbar \overline{y_{N,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{N,i_2,r_1} Y_{e,i_3,i_1} \\
& + \frac{\hbar \overline{y_{N,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_\eta^2 y_{N,i_2,r_1} Y_{e,i_3,i_1}}{2 (M_\eta^2 - M_N^2)} \\
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_\eta^2} \right) \overline{y_{N,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_\eta^2 y_{N,i_2,r_1} Y_{e,i_3,i_1}}{2 (M_\eta^2 - M_N^2)} \\
& - \frac{\hbar \overline{y_{N,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_\eta^4 y_{N,i_2,r_1} Y_{e,i_3,i_1}}{4 (M_\eta^2 - M_N^2)^2} \\
& - \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_\eta^2} \right) \overline{y_{N,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_\eta^4 y_{N,i_2,r_1} Y_{e,i_3,i_1}}{4 (M_\eta^2 - M_N^2)^2} \\
& + C_{H2} H_{a_1} H_{a_1}^\dagger \\
& + \frac{\hbar \overline{y_{N,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{N,i_3,r_1} H_{a_1}^\dagger}{4 \epsilon} \\
& - \frac{1}{8} \hbar \overline{y_{N,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{N,i_3,r_1} H_{a_1}^\dagger \\
& + \frac{\hbar \overline{y_{N,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_\eta^2 y_{N,i_3,r_1} H_{a_1}^\dagger}{2 (M_\eta^2 - M_N^2)}
\end{aligned}$$

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$$\begin{aligned}
& + \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_\eta^2} \right) \overline{y_{N,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_\eta^2 y_{N,i_3,r_1} H_{a_1}^\dagger}{2 (M_\eta^2 - M_N^2)} \\
& - \frac{\hbar \overline{y_{N,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_\eta^4 y_{N,i_3,r_1} H_{a_1}^\dagger}{4 (M_\eta^2 - M_N^2)^2} \\
& - \frac{\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_\eta^2} \right) \overline{y_{N,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_\eta^4 y_{N,i_3,r_1} H_{a_1}^\dagger}{4 (M_\eta^2 - M_N^2)^2} \\
& - H_{a_1} \mu^2 H_{a_1}^\dagger \\
& + \frac{\hbar H_{a_1} H_{a_2} \lambda_3^2 H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\
& + \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_\eta^2} \right) H_{a_1} H_{a_2} \lambda_3^2 H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{\hbar H_{a_1} H_{a_2} \lambda_4^2 H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& + \frac{1}{2} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_\eta^2} \right) H_{a_1} H_{a_2} \lambda_4^2 H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{\hbar H_{a_1} H_{a_2} \lambda_3 \lambda_4 H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\
& + \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_\eta^2} \right) H_{a_1} H_{a_2} \lambda_3 \lambda_4 H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{\hbar \overline{\lambda_5} H_{a_1} H_{a_2} \lambda_5 H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& + \frac{1}{2} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_\eta^2} \right) \overline{\lambda_5} H_{a_1} H_{a_2} \lambda_5 H_{a_1}^\dagger H_{a_2}^\dagger
\end{aligned}$$

$$d_S = 2, \quad Y_S = \frac{1}{2}, \quad d_F = 3, \quad Y_F = 0$$

$$\begin{aligned}
& \textbf{BSM UV Lagrangian} \quad \eta_{a_2} \bar{\epsilon}_{a_1,a_2} \overline{y_{N,i_1,r_1} \text{gen}(\text{SU2L(fund)})}_{A_1,a_1,a_3} N_{A_1,r_1}^T C P_L \ell_{a_3,i_1} \\
& + \epsilon_{a_2,a_1} \eta_{a_1}^\dagger y_{N,i_1,r_1} \text{gen}(\text{SU2L(fund)})_{A_1,a_3,a_2} \overline{\ell_{a_3,i_1}} P_R N_{A_1,r_1} \\
& + \frac{1}{2} \overline{\lambda_5} H_{a_1} H_{a_2} \eta_{a_1}^\dagger \eta_{a_2}^\dagger \\
& + \frac{1}{2} i N_{A_1,r_1}^T C \gamma^{\mu_1} D_{\mu_1} (N_{A_1,r_1}) \\
& - \frac{1}{2} M_N N_{A_1,r_1}^T C N_{A_1,r_1} \\
& + \frac{1}{2} \lambda_5 \eta_{a_1} \eta_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \lambda_3 H_{a_1} \eta_{a_2} H_{a_1}^\dagger \eta_{a_2}^\dagger \\
& + \lambda_4 \eta_{a_1} H_{a_2} H_{a_1}^\dagger \eta_{a_2}^\dagger \\
& + \frac{1}{2} \lambda_S \eta_{a_1} \eta_{a_2} \eta_{a_1}^\dagger \eta_{a_2}^\dagger \\
& + D_{\mu_1} (\eta_{a_1}) D_{\mu_1} (\eta_{a_1})^\dagger
\end{aligned}$$

$$\begin{aligned}
& + M_\eta^2 (-\eta_{a_1}) \eta_{a_1}^\dagger \\
& \Delta \mathcal{L}_{\text{EFT}} - \frac{3\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_N^2}\right) \overline{y_{N,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{N,i_2,r_1} Y_{e,i_3,i_1} M_N^4}{16 (M_\eta^2 - M_N^2)^2} \\
& - \frac{3\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_N^2}\right) \overline{y_{N,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{N,i_3,r_1} H_{a_1}^\dagger M_N^4}{16 (M_\eta^2 - M_N^2)^2} \\
& + \frac{\hbar \overline{y_{N,i_1,r_1} y_{N,i_2,r_1} \lambda_5 \ell_{a_3,i_1}}^T C P_L \ell_{a_2,i_2} H_{a_1} H_{a_4} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_N^3}{8 (M_\eta^2 - M_N^2)^2} \\
& + \frac{\hbar \log\left(\frac{M_\eta^2}{M_N^2}\right) \overline{y_{N,i_1,r_1} y_{N,i_2,r_1} \lambda_5 \ell_{a_3,i_1}}^T C P_L \ell_{a_2,i_2} H_{a_1} H_{a_4} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_N^3}{8 (M_\eta^2 - M_N^2)^2} \\
& + \frac{\hbar \overline{\ell_{a_3,i_2}} C P_R \overline{\ell_{a_4,i_1}}^T y_{N,i_1,r_1} y_{N,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_4,a_2} \lambda_5 H_{a_1}^\dagger H_{a_2}^\dagger M_N^3}{8 (M_\eta^2 - M_N^2)^2} \\
& + \frac{\hbar \log\left(\frac{M_\eta^2}{M_N^2}\right) \overline{\ell_{a_3,i_2}} C P_R \overline{\ell_{a_4,i_1}}^T y_{N,i_1,r_1} y_{N,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_4,a_2} \lambda_5 H_{a_1}^\dagger H_{a_2}^\dagger M_N^3}{8 (M_\eta^2 - M_N^2)^2} \\
& - \frac{3\hbar \overline{y_{N,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{N,i_2,r_1} Y_{e,i_3,i_1} M_N^2}{8 (M_\eta^2 - M_N^2)} \\
& - \frac{3\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_N^2}\right) \overline{y_{N,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{N,i_2,r_1} Y_{e,i_3,i_1} M_N^2}{8 (M_\eta^2 - M_N^2)} \\
& + \frac{3\hbar \overline{y_{N,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_\eta^2 y_{N,i_2,r_1} Y_{e,i_3,i_1} M_N^2}{16 (M_\eta^2 - M_N^2)^2} \\
& + \frac{3\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_N^2}\right) \overline{y_{N,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_\eta^2 y_{N,i_2,r_1} Y_{e,i_3,i_1} M_N^2}{8 (M_\eta^2 - M_N^2)^2} \\
& - \frac{3\hbar \overline{y_{N,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{N,i_3,r_1} H_{a_1}^\dagger M_N^2}{8 (M_\eta^2 - M_N^2)} \\
& - \frac{3\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_N^2}\right) \overline{y_{N,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{N,i_3,r_1} H_{a_1}^\dagger M_N^2}{8 (M_\eta^2 - M_N^2)} \\
& + \frac{3\hbar \overline{y_{N,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_\eta^2 y_{N,i_3,r_1} H_{a_1}^\dagger M_N^2}{16 (M_\eta^2 - M_N^2)^2} \\
& + \frac{3\hbar \log\left(\frac{\mu_{\text{bar}2}}{M_N^2}\right) \overline{y_{N,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_\eta^2 y_{N,i_3,r_1} H_{a_1}^\dagger M_N^2}{8 (M_\eta^2 - M_N^2)^2} \\
& - \frac{\hbar \overline{y_{N,i_1,r_1} y_{N,i_2,r_1} \lambda_5 \ell_{a_3,i_1}}^T C P_L \ell_{a_2,i_2} H_{a_1} H_{a_4} M_\eta^2 \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_N}{8 (M_\eta^2 - M_N^2)^2} \\
& - \frac{\hbar \overline{\ell_{a_3,i_2}} C P_R \overline{\ell_{a_4,i_1}}^T M_\eta^2 y_{N,i_1,r_1} y_{N,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_4,a_2} \lambda_5 H_{a_1}^\dagger H_{a_2}^\dagger M_N}{8 (M_\eta^2 - M_N^2)^2} \\
& - \frac{\hbar B_{\mu_1 \mu_2}^2}{24\epsilon}
\end{aligned}$$

$$\begin{aligned}
& -\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)B_{\mu_1\mu_2}^2 \\
& -\frac{25\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{24\epsilon} \\
& -\hbar\log\left(\frac{\mu\text{bar}2}{M_N^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2 \\
& -\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2 \\
& +\frac{3\hbar\overline{y_{N,i_3,r_1}\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{N,i_2,r_1}Y_{e,i_3,i_1}}{16\epsilon} \\
& -\frac{3}{32}\hbar\overline{y_{N,i_3,r_1}\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{N,i_2,r_1}Y_{e,i_3,i_1} \\
& +\frac{3\hbar\overline{y_{N,i_3,r_1}\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_\eta^2y_{N,i_2,r_1}Y_{e,i_3,i_1}}{8\left(M_\eta^2-M_N^2\right)} \\
& +\frac{3\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)\overline{y_{N,i_3,r_1}\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_\eta^2y_{N,i_2,r_1}Y_{e,i_3,i_1}}{8\left(M_\eta^2-M_N^2\right)} \\
& -\frac{3\hbar\overline{y_{N,i_3,r_1}\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_\eta^4y_{N,i_2,r_1}Y_{e,i_3,i_1}}{16\left(M_\eta^2-M_N^2\right)^2} \\
& -\frac{3\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)\overline{y_{N,i_3,r_1}\ell_{a_1,i_2}}P_R e_{i_1}H_{a_1}M_\eta^4y_{N,i_2,r_1}Y_{e,i_3,i_1}}{16\left(M_\eta^2-M_N^2\right)^2} \\
& +C_{H2}H_{a_1}H_{a_1}^\dagger \\
& +\frac{3\hbar\overline{y_{N,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L\ell_{a_1,i_1}y_{N,i_3,r_1}H_{a_1}^\dagger}{16\epsilon} \\
& -\frac{3}{32}\hbar\overline{y_{N,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L\ell_{a_1,i_1}y_{N,i_3,r_1}H_{a_1}^\dagger \\
& +\frac{3\hbar\overline{y_{N,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L\ell_{a_1,i_1}M_\eta^2y_{N,i_3,r_1}H_{a_1}^\dagger}{8\left(M_\eta^2-M_N^2\right)} \\
& +\frac{3\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)\overline{y_{N,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L\ell_{a_1,i_1}M_\eta^2y_{N,i_3,r_1}H_{a_1}^\dagger}{8\left(M_\eta^2-M_N^2\right)} \\
& -\frac{3\hbar\overline{y_{N,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L\ell_{a_1,i_1}M_\eta^4y_{N,i_3,r_1}H_{a_1}^\dagger}{16\left(M_\eta^2-M_N^2\right)^2} \\
& -\frac{3\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)\overline{y_{N,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L\ell_{a_1,i_1}M_\eta^4y_{N,i_3,r_1}H_{a_1}^\dagger}{16\left(M_\eta^2-M_N^2\right)^2} \\
& -H_{a_1}\mu^2H_{a_1}^\dagger \\
& +\frac{\hbar H_{a_1}H_{a_2}\lambda_3^2H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\
& +\hbar\log\left(\frac{\mu\text{bar}2}{M_\eta^2}\right)H_{a_1}H_{a_2}\lambda_3^2H_{a_1}^\dagger H_{a_2}^\dagger
\end{aligned}$$

$$\begin{aligned}
& + \frac{\hbar H_{a_1} H_{a_2} \lambda_4^2 H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& + \frac{1}{2} \hbar \log \left(\frac{\mu_{\text{bar}2}}{M_\eta^2} \right) H_{a_1} H_{a_2} \lambda_4^2 H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{\hbar H_{a_1} H_{a_2} \lambda_3 \lambda_4 H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\
& + \hbar \log \left(\frac{\mu_{\text{bar}2}}{M_\eta^2} \right) H_{a_1} H_{a_2} \lambda_3 \lambda_4 H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{\hbar \overline{\lambda_5} H_{a_1} H_{a_2} \lambda_5 H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& + \frac{1}{2} \hbar \log \left(\frac{\mu_{\text{bar}2}}{M_\eta^2} \right) \overline{\lambda_5} H_{a_1} H_{a_2} \lambda_5 H_{a_1}^\dagger H_{a_2}^\dagger
\end{aligned}$$

$$d_S = 3, \quad Y_S = \frac{1}{2}, \quad d_F = 2, \quad Y_F = 0$$

$$\begin{aligned}
& \textbf{BSM UV Lagrangian } \eta_{A_1} \overline{y_{N,i_1,r_1}} \text{gen}(\text{SU2L(fund)})_{A_1,a_1,a_2} \overline{N_{a_1,r_1}} P_L \ell_{a_2,i_1} \\
& + \eta_{A_1}^\dagger y_{N,i_1,r_1} \text{gen}(\text{SU2L(fund)})_{A_1,a_1,a_2} \overline{\ell_{a_1,i_1}} P_R N_{a_2,r_1} \\
& + i \overline{N_{a_1,r_1}} \gamma^{\mu_1} D_{\mu_1} (N_{a_1,r_1}) \\
& - M_N \overline{N_{a_1,r_1}} N_{a_1,r_1} \\
& + \frac{1}{2} \lambda_S \eta_{A_1} \eta_{A_2} \eta_{A_1}^\dagger \eta_{A_2}^\dagger \\
& + \lambda_3 H_{a_1} \eta_{A_1} H_{a_1}^\dagger \eta_{A_1}^\dagger \\
& + D_{\mu_1} (\eta_{A_1}) D_{\mu_1} (\eta_{A_1})^\dagger \\
& + M_\eta^2 (-\eta_{A_1}) \eta_{A_1}^\dagger \\
& \Delta \mathcal{L}_{\text{EFT}} - \frac{3 \hbar \log \left(\frac{\mu_{\text{bar}2}}{M_N^2} \right) \overline{y_{N,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{N,i_2,r_1} Y_{e,i_3,i_1} M_N^4}{16 (M_\eta^2 - M_N^2)^2} \\
& - \frac{3 \hbar \log \left(\frac{\mu_{\text{bar}2}}{M_N^2} \right) \overline{y_{N,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{N,i_3,r_1} H_{a_1}^\dagger M_N^4}{16 (M_\eta^2 - M_N^2)^2} \\
& - \frac{3 \hbar \overline{y_{N,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{N,i_2,r_1} Y_{e,i_3,i_1} M_N^2}{8 (M_\eta^2 - M_N^2)} \\
& - \frac{3 \hbar \log \left(\frac{\mu_{\text{bar}2}}{M_N^2} \right) \overline{y_{N,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{N,i_2,r_1} Y_{e,i_3,i_1} M_N^2}{8 (M_\eta^2 - M_N^2)} \\
& + \frac{3 \hbar \overline{y_{N,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_\eta^2 y_{N,i_2,r_1} Y_{e,i_3,i_1} M_N^2}{16 (M_\eta^2 - M_N^2)^2} \\
& + \frac{3 \hbar \log \left(\frac{\mu_{\text{bar}2}}{M_N^2} \right) \overline{y_{N,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_\eta^2 y_{N,i_2,r_1} Y_{e,i_3,i_1} M_N^2}{8 (M_\eta^2 - M_N^2)^2} \\
& - \frac{3 \hbar \overline{y_{N,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{N,i_3,r_1} H_{a_1}^\dagger M_N^2}{8 (M_\eta^2 - M_N^2)}
\end{aligned}$$

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$$\begin{aligned}
& - \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_N^2} \right) \overline{y_{N,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{N,i_3,r_1} H_{a_1}{}^\dagger M_N^2}{8 (M_\eta^2 - M_N^2)} \\
& + \frac{3\hbar \overline{y_{N,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_\eta^2 y_{N,i_3,r_1} H_{a_1}{}^\dagger M_N^2}{16 (M_\eta^2 - M_N^2)^2} \\
& + \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_N^2} \right) \overline{y_{N,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_\eta^2 y_{N,i_3,r_1} H_{a_1}{}^\dagger M_N^2}{8 (M_\eta^2 - M_N^2)^2} \\
& - \frac{\hbar B_{\mu_1 \mu_2}^2}{16\epsilon} \\
& - \frac{1}{16} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_\eta^2} \right) B_{\mu_1 \mu_2}^2 \\
& - \frac{2\hbar (W_{\mu_1 \mu_2}^{A_1})^2}{3\epsilon} \\
& - \frac{1}{2} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_N^2} \right) (W_{\mu_1 \mu_2}^{A_1})^2 \\
& - \frac{1}{6} \hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_\eta^2} \right) (W_{\mu_1 \mu_2}^{A_1})^2 \\
& + \frac{3\hbar \overline{y_{N,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{N,i_2,r_1} Y_{e,i_3,i_1}}{16\epsilon} \\
& - \frac{3}{32} \hbar \overline{y_{N,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{N,i_2,r_1} Y_{e,i_3,i_1} \\
& + \frac{3\hbar \overline{y_{N,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_\eta^2 y_{N,i_2,r_1} Y_{e,i_3,i_1}}{8 (M_\eta^2 - M_N^2)} \\
& + \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_\eta^2} \right) \overline{y_{N,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_\eta^2 y_{N,i_2,r_1} Y_{e,i_3,i_1}}{8 (M_\eta^2 - M_N^2)} \\
& - \frac{3\hbar \overline{y_{N,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_\eta^4 y_{N,i_2,r_1} Y_{e,i_3,i_1}}{16 (M_\eta^2 - M_N^2)^2} \\
& - \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_\eta^2} \right) \overline{y_{N,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_\eta^4 y_{N,i_2,r_1} Y_{e,i_3,i_1}}{16 (M_\eta^2 - M_N^2)^2} \\
& + C_{H2} H_{a_1} H_{a_1}{}^\dagger \\
& + \frac{3\hbar \overline{y_{N,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{N,i_3,r_1} H_{a_1}{}^\dagger}{16\epsilon} \\
& - \frac{3}{32} \hbar \overline{y_{N,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} y_{N,i_3,r_1} H_{a_1}{}^\dagger \\
& + \frac{3\hbar \overline{y_{N,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_\eta^2 y_{N,i_3,r_1} H_{a_1}{}^\dagger}{8 (M_\eta^2 - M_N^2)} \\
& + \frac{3\hbar \log \left(\frac{\mu_{\text{bar}}^2}{M_\eta^2} \right) \overline{y_{N,i_1,r_1} Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_\eta^2 y_{N,i_3,r_1} H_{a_1}{}^\dagger}{8 (M_\eta^2 - M_N^2)}
\end{aligned}$$

$$\begin{aligned}
& - \frac{3\hbar \overline{y_{N,i_1,r_1}} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_\eta^4 y_{N,i_3,r_1} H_{a_1}^\dagger}{16 (M_\eta^2 - M_N^2)^2} \\
& - \frac{3\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_\eta^2}\right) \overline{y_{N,i_1,r_1}} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L \ell_{a_1,i_1} M_\eta^4 y_{N,i_3,r_1} H_{a_1}^\dagger}{16 (M_\eta^2 - M_N^2)^2} \\
& - H_{a_1} \mu^2 H_{a_1}^\dagger \\
& + \frac{3\hbar H_{a_1} H_{a_2} \lambda_3^2 H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\
& + \frac{3}{2} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_\eta^2}\right) H_{a_1} H_{a_2} \lambda_3^2 H_{a_1}^\dagger H_{a_2}^\dagger
\end{aligned}

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